

Problem 16.6

Megan wants to buy a car in 4 years for \$18,000. She deposits X at the beginning of each month for four years into an account earning 6% compounded monthly. Calculate X .

Solution

The nominal annual interest rate compounded monthly is 6%, so the effective monthly interest rate is

$$j = \frac{0.06}{12} = 0.005.$$

There are

$$4(12) = 48$$

monthly deposits. Since the deposits are made at the beginning of each month, this is an annuity due. The accumulated value at the end of four years is

$$X\ddot{s}_{\overline{48}|} = 18000.$$

For an annuity due,

$$\ddot{s}_{\overline{n}|} = (1 + j)s_{\overline{n}|}.$$

Therefore,

$$\ddot{s}_{\overline{48}|} = (1.005) \left(\frac{(1.005)^{48} - 1}{0.005} \right).$$

Thus,

$$X(1.005) \left(\frac{(1.005)^{48} - 1}{0.005} \right) = 18000.$$

Solving for X ,

$$\begin{aligned} X &= \frac{18000}{(1.005) \left(\frac{(1.005)^{48} - 1}{0.005} \right)} \\ &\approx 331.0751. \end{aligned}$$

Hence, the monthly deposit is

$$\boxed{X \approx \$331.08}.$$